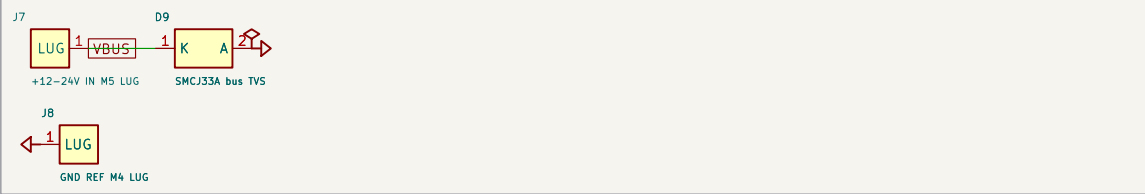
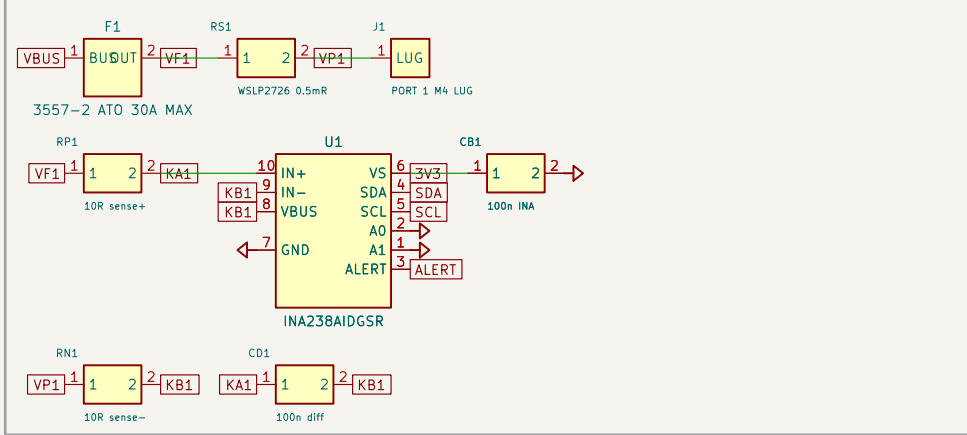


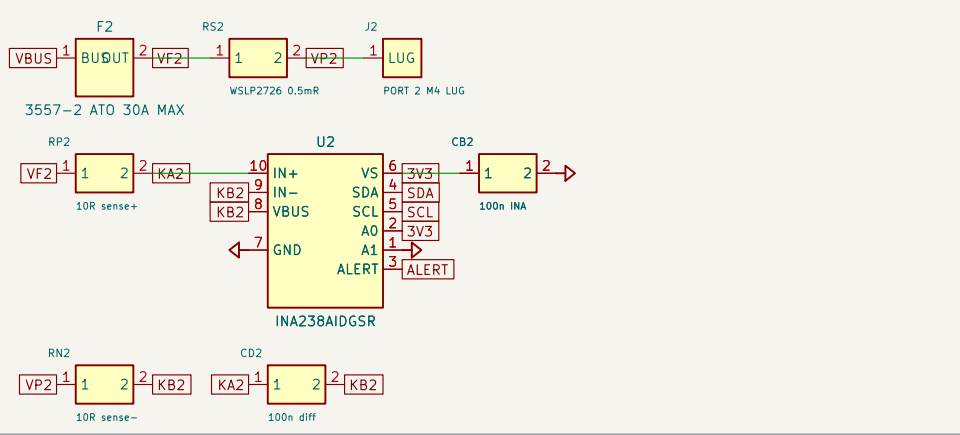
1. BUS INPUT: M5 lug 60A + SMCJ33A clamp; GND is REFERENCE only (D1) [ADR-0001/0005]



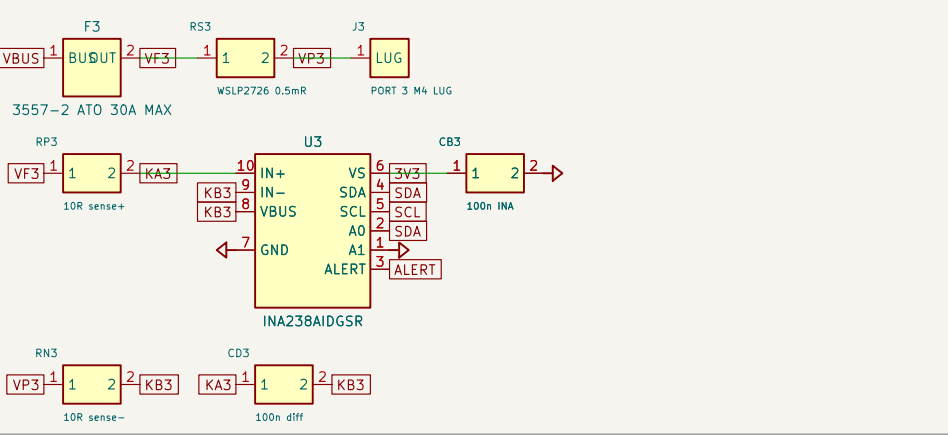
2.1 PORT 1: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x40 [ADR-0003]



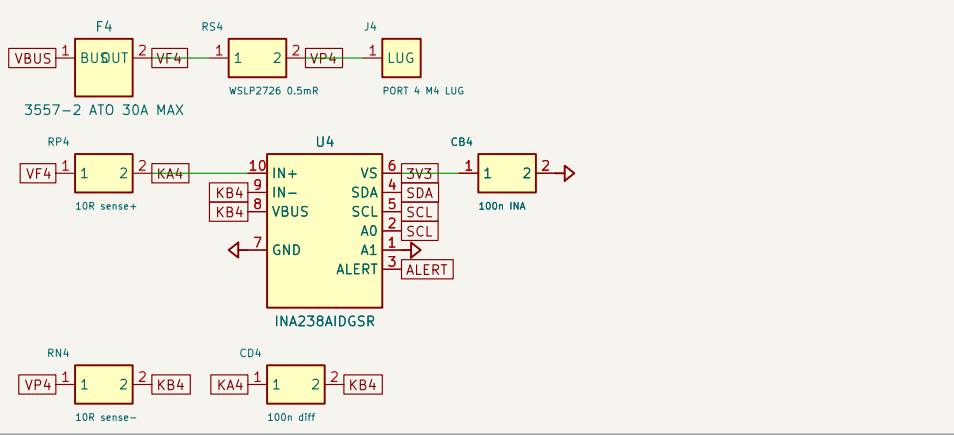
2.2 PORT 2: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x41 [ADR-0003]



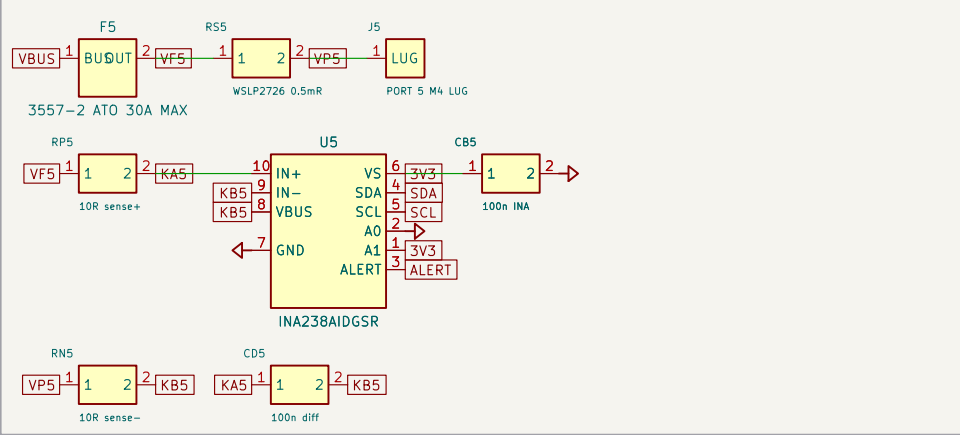
2.3 PORT 3: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x42 [ADR-0003]



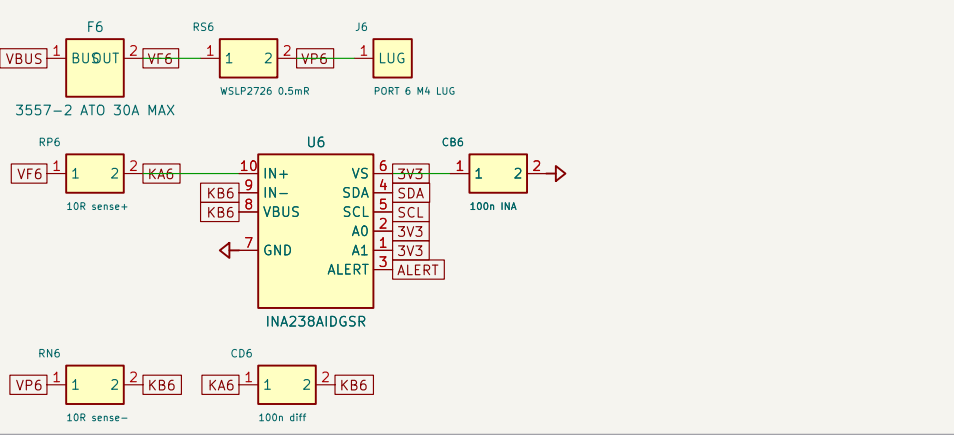
2.4 PORT 4: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x43 [ADR-0003]



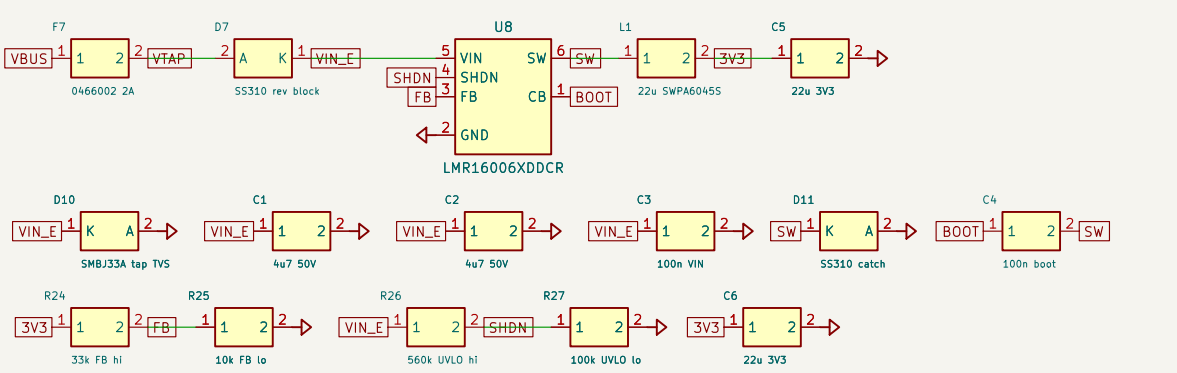
2.5 PORT 5: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x44 [ADR-0003]



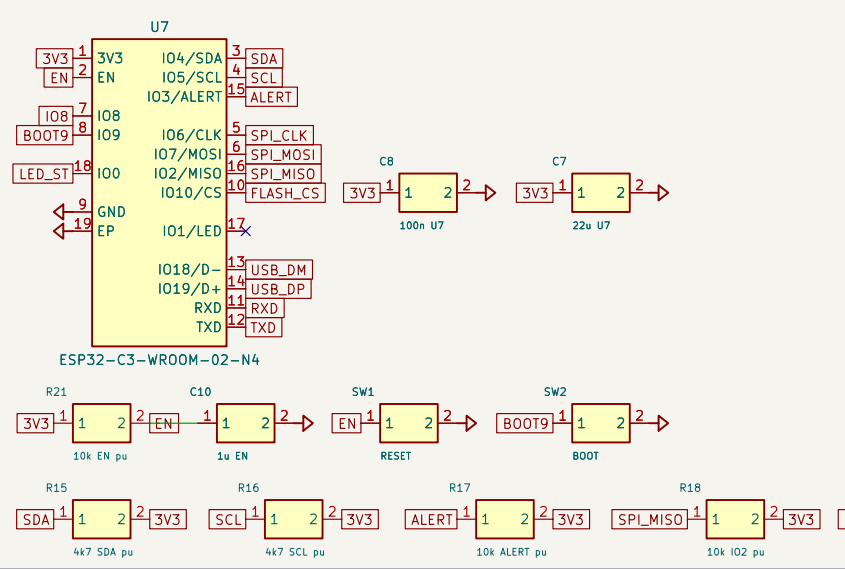
2.6 PORT 6: ATO fuse -> 0.5m shunt -> M4 stud; INA238 @0x45 [ADR-0003]



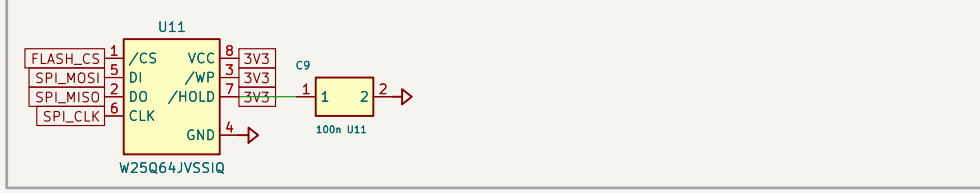
3. ELECTRONICS: 2A fuse -> reverse-block -> LMR16006X 60V buck -> 3.3V [ADR-0001/0006]



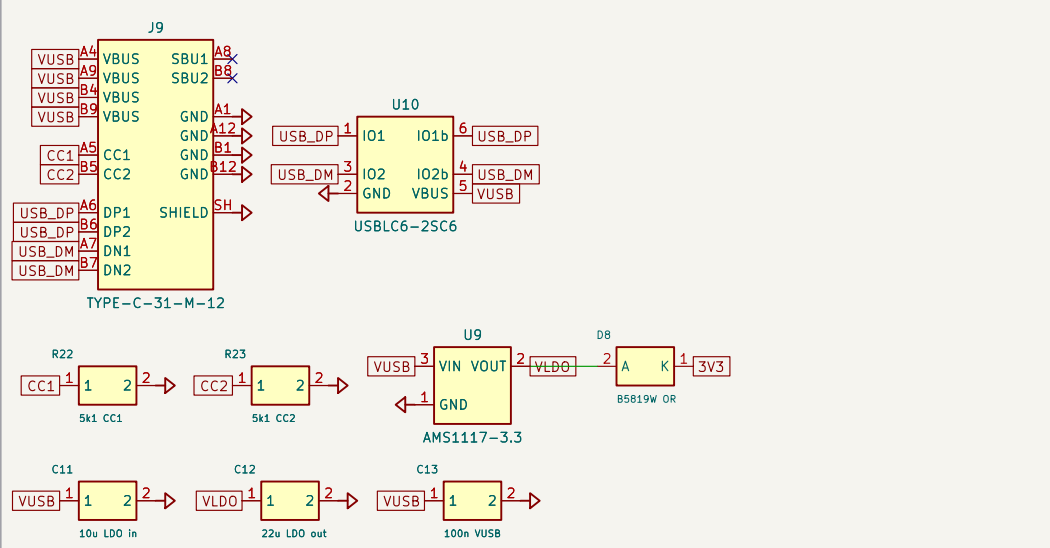
4. ESP32-C3-WROOM-02: BLE 5.0 + native USB; straps per ADR-0004



5. LOG FLASH: W25Q64JV 8MB stats ring (dedicated, not app flash) [ADR-0004]



6. USB-C: flash/debug + bench power (AMS1117 -> B5819W OR, -3.0V) [ADR-0006]



7. LEDS + DEBUG UART (J10 DNP)

